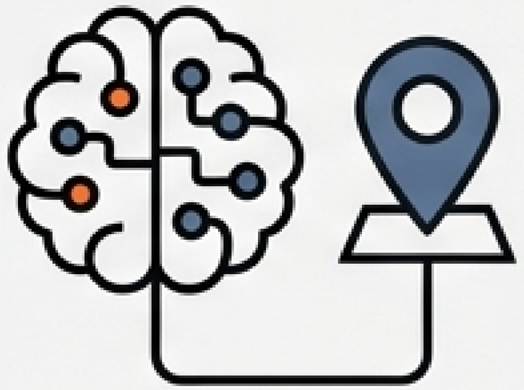
Road Accident Hotspot Detection & Risk Prediction

Transitioning urban safety from reactive manual analysis to proactive AI foresight.

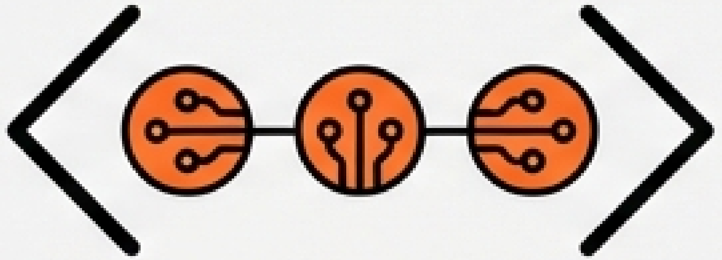
A Project by Team BACK | Domain: AI/ML

Executive Summary

The Challenge: High accident rates in urban areas coupled with a lack of predictive systems.



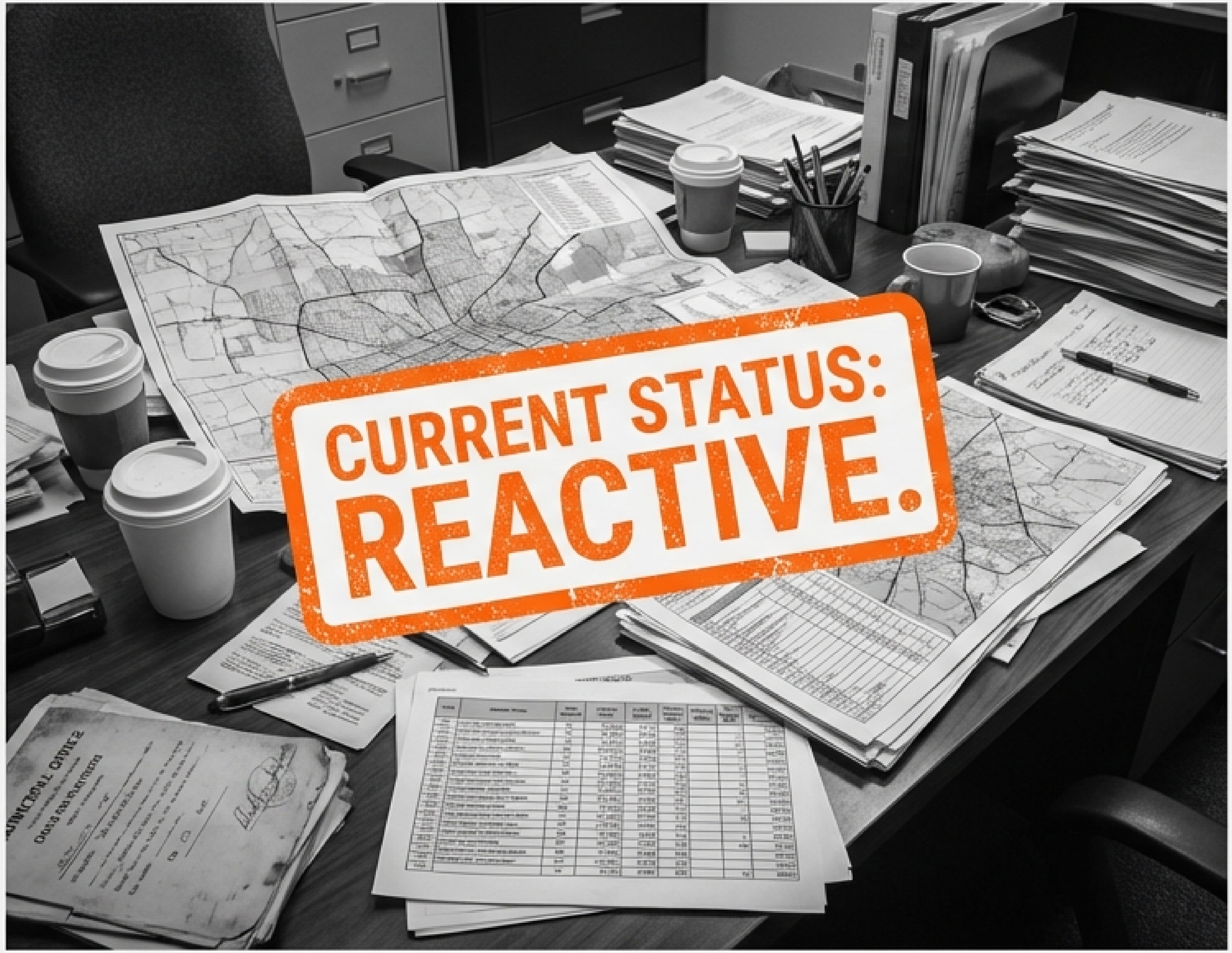
The Solution: An AI-ecosystem that detects spatial hotspots and predicts risk scores using historical data.



The Core Tech: Unsupervised Clustering (Hotspots) + Supervised Regression (Risk) + Interactive Dashboard.

The Outcome: Data-driven preventive planning for traffic authorities and improved public safety.

The Reactive Trap of Manual Analysis



Connection Timed Out

Authorities currently rely on historical and manual analysis. This approach is retrospective—investigating accidents only after they have occurred.

- Focus limited to post-event reporting.
- Inability to process complex variables for prediction.
- Crucial time lag between accident trends and safety interventions.

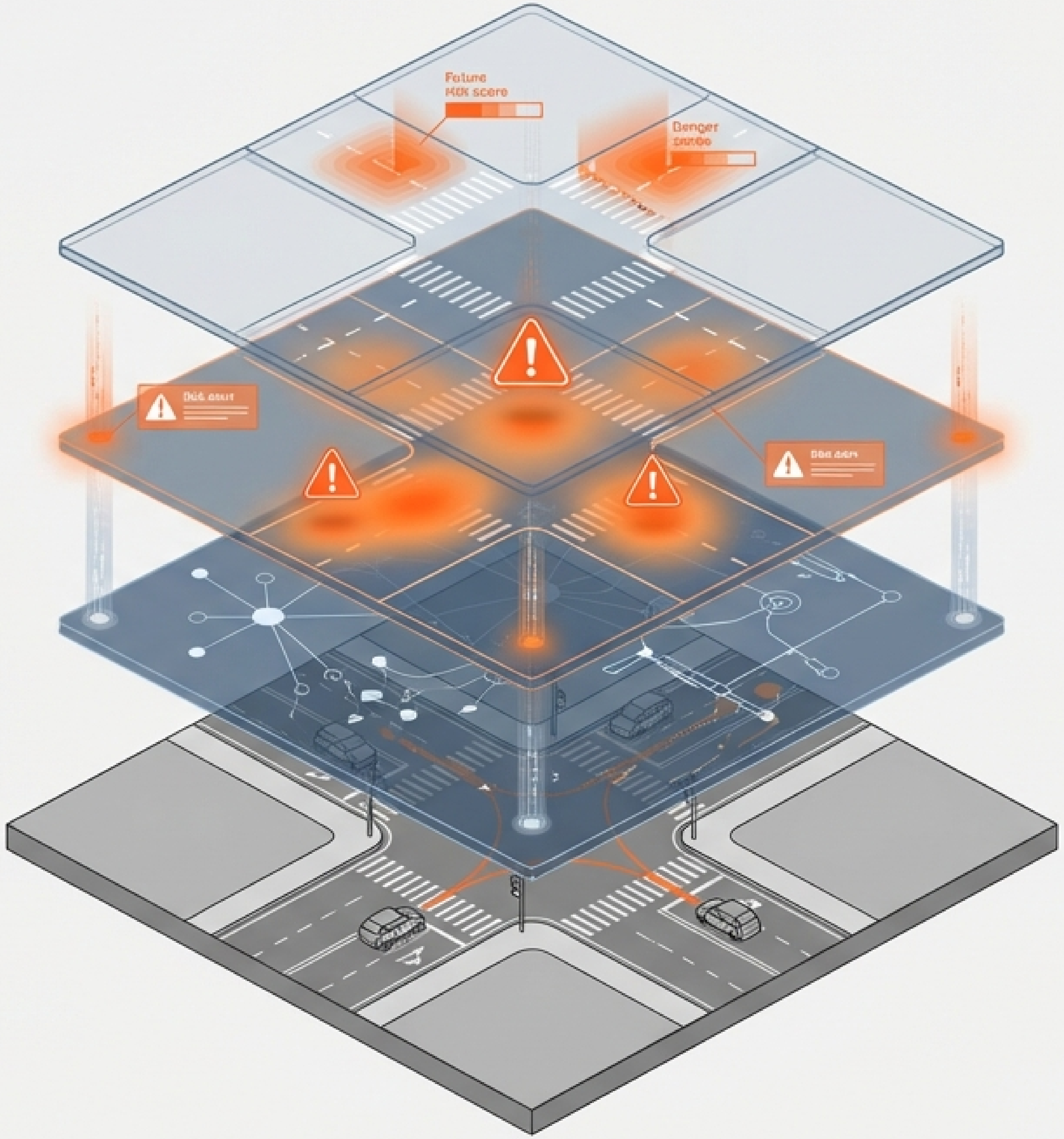
A Proactive Approach to Risk

We propose a shift from “**What happened?**” to “**What might happen?**”. Our system leverages AI to identify accident-prone zones before they become statistics.

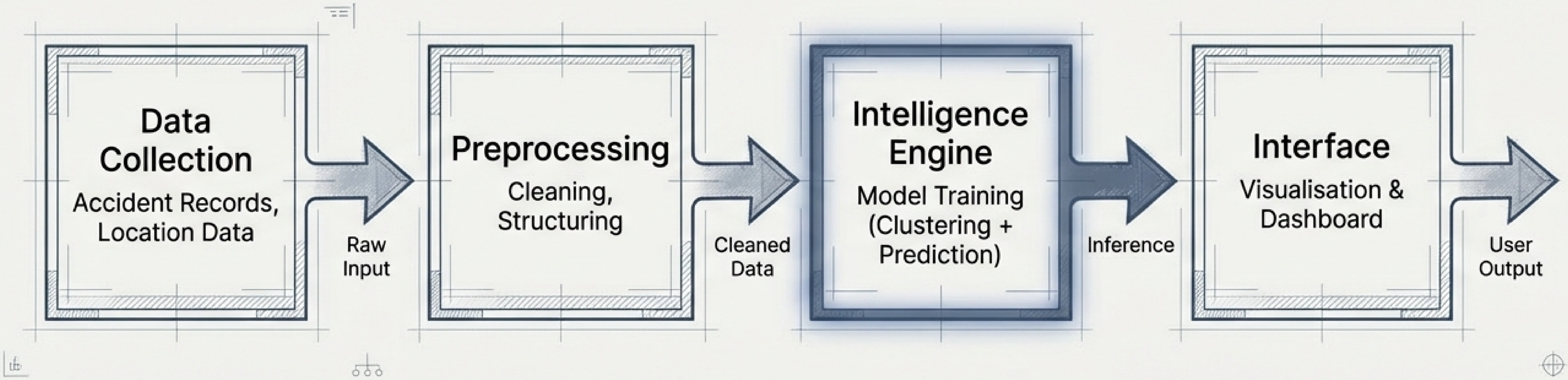
Detect: Pinpoint high-density accident zones.

Predict: Calculate risk probability based on environmental variables.

Visualise: Deliver actionable insights via heatmaps.



System Architecture Overview



The Data Foundation

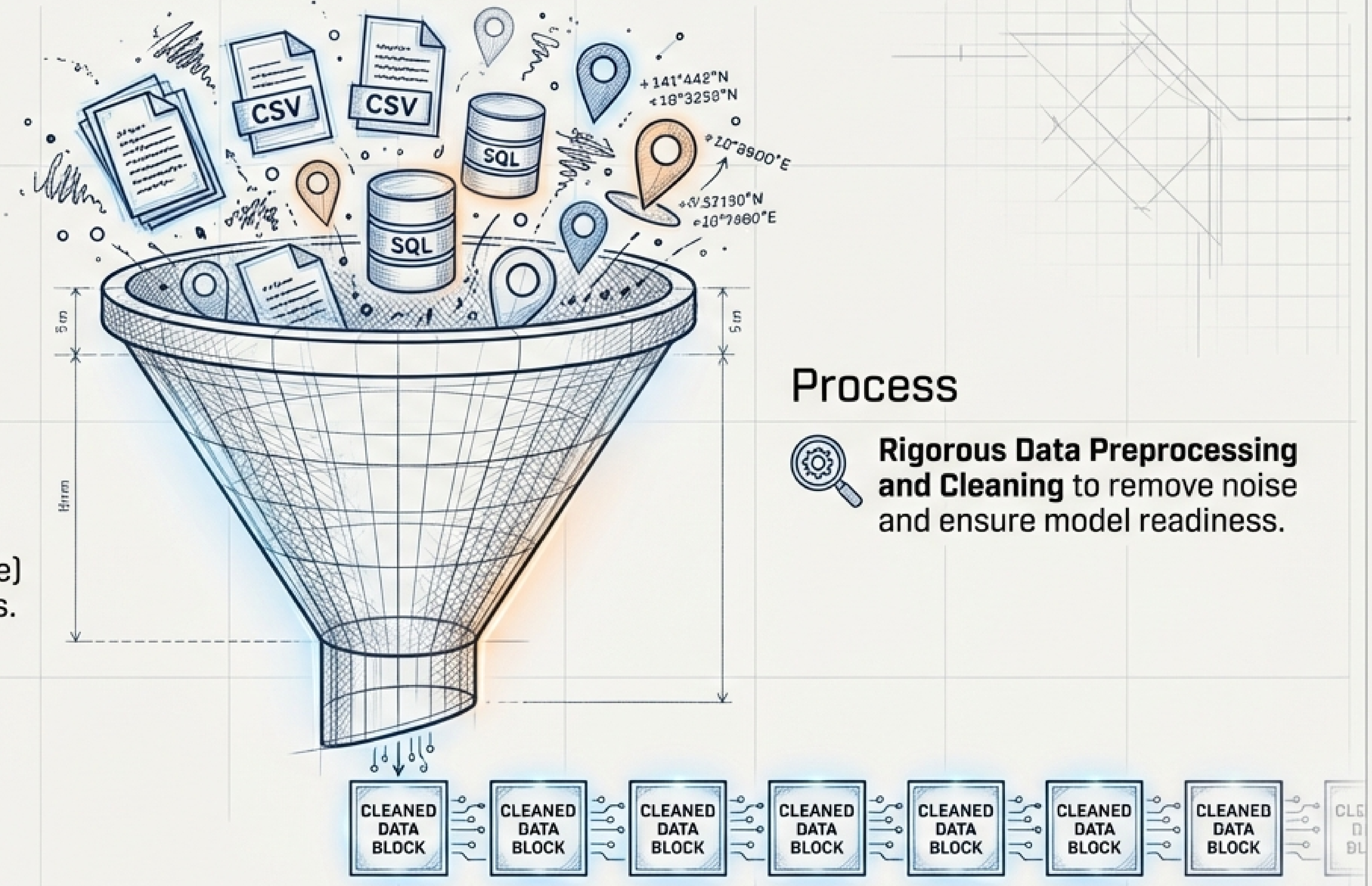
Input Sources

Accident Records: Historical logs of past incidents.

Location Data: Geospatial coordinates (latitude/longitude) of urban and semi-urban areas.

Process

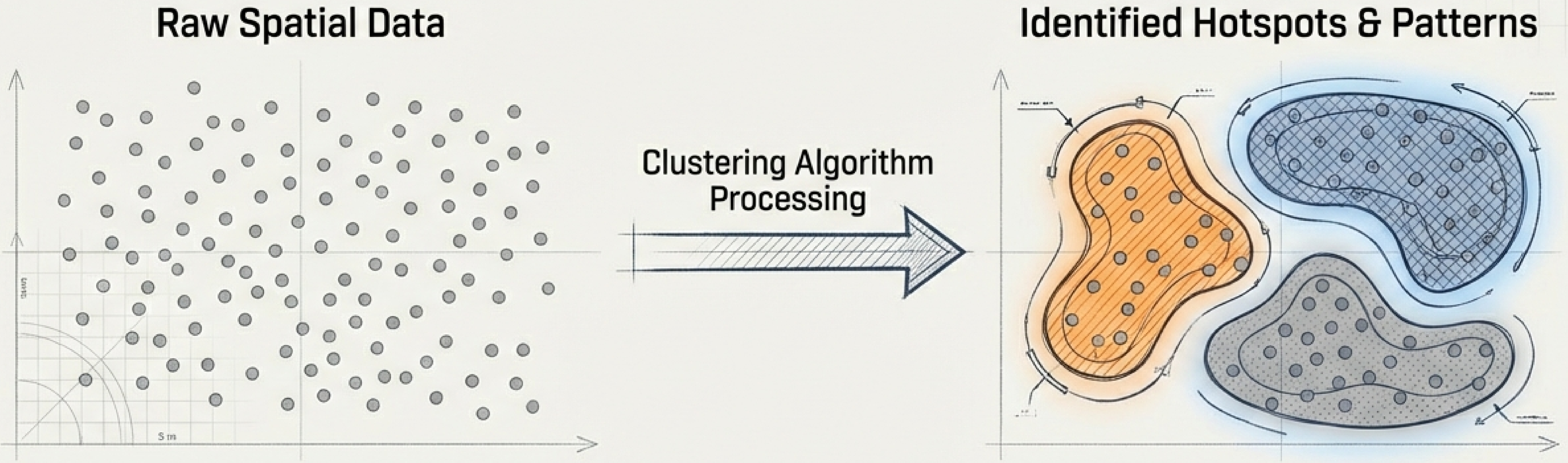
Rigorous Data Preprocessing and Cleaning to remove noise and ensure model readiness.



Structured Data for Model Readiness

Mechanism I: Hotspot Detection

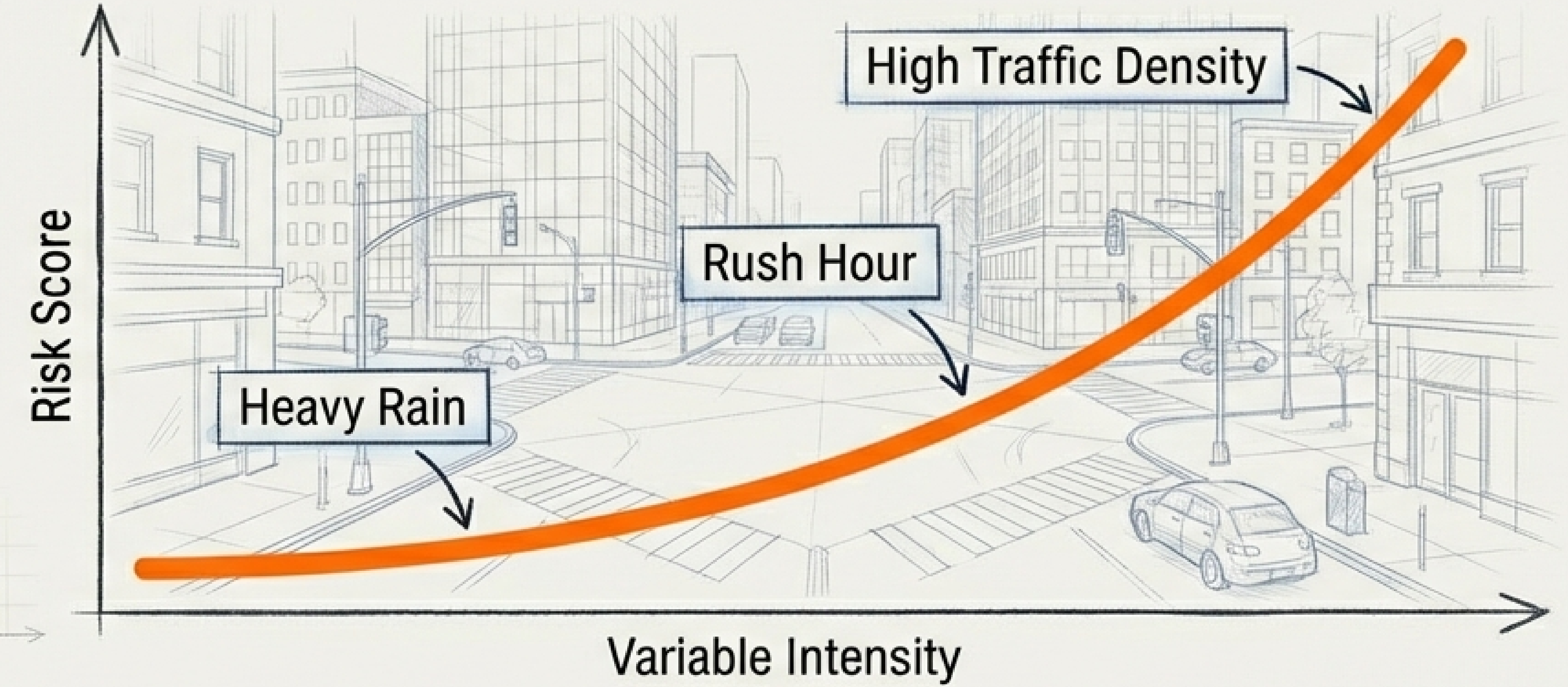
Tech Highlight: Unsupervised Learning



- **Algorithm**: Utilising Clustering algorithms such as K-Means or DBSCAN.
- **Function**: The system aggregates spatial data points to mathematically define 'accident-prone zones,' removing human bias from the identification process.

Mechanism II: Risk Prediction

Tech Highlight: Supervised Learning



Algorithm: Deployment of Regression and Classification models.

Feature Engineering: The model analyses specific variables to calculate risk scores:

- Weather conditions
- Time of day
- Traffic density

Technology Stack

Core Logic

Python

Machine Learning

Scikit-learn
TensorFlow

Visualisation

Matplotlib
Plotly

Data

SQL
CSV

Deployment

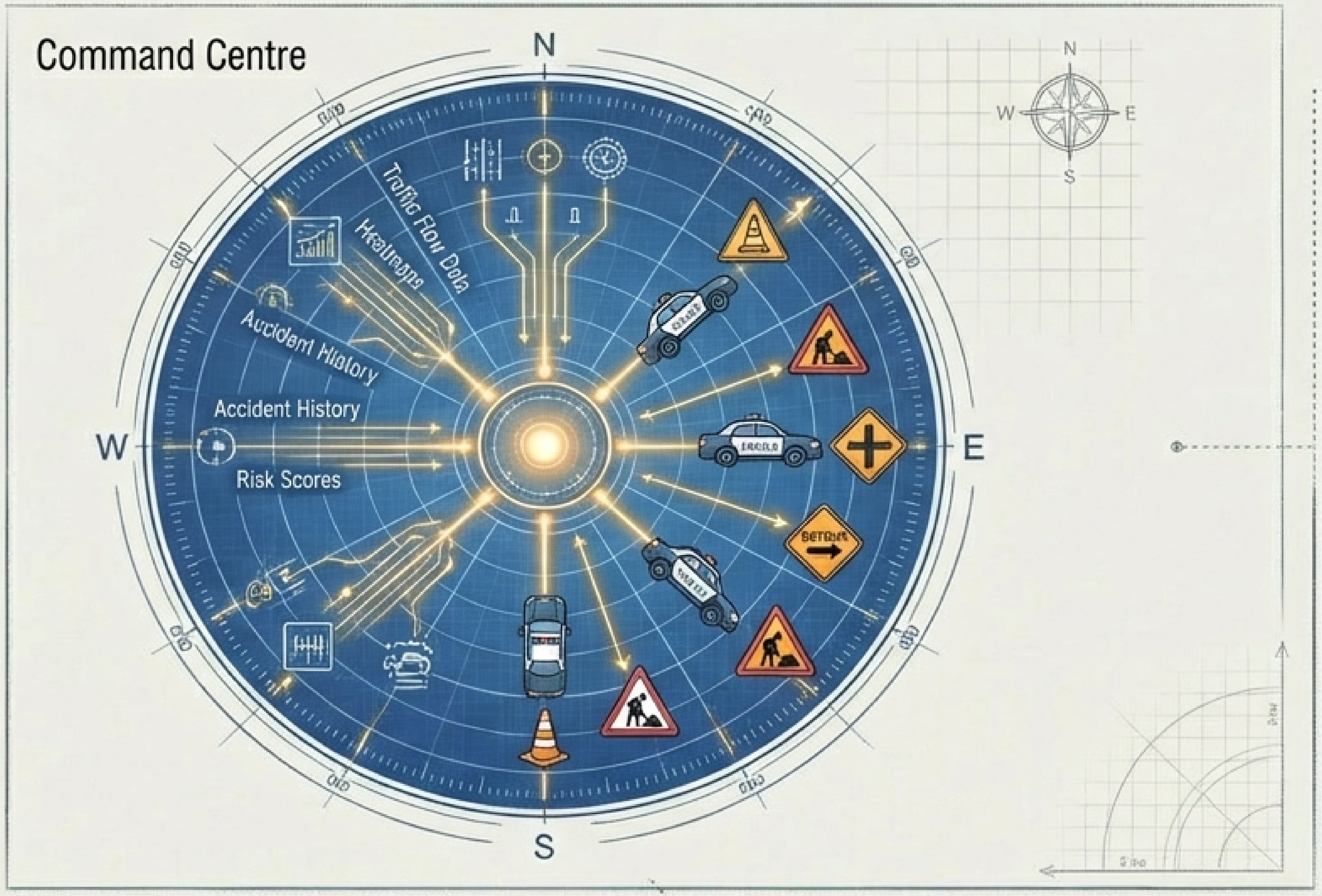
Streamlit
Flask

The User Experience

- **Feature: Interactive Map Visualisation**

Authorities access a Web Dashboard (built on Streamlit/Flask) that renders complex data as intuitive visual heatmaps and risk scores, **allowing for immediate situational awareness.**

Empowering Traffic Authorities

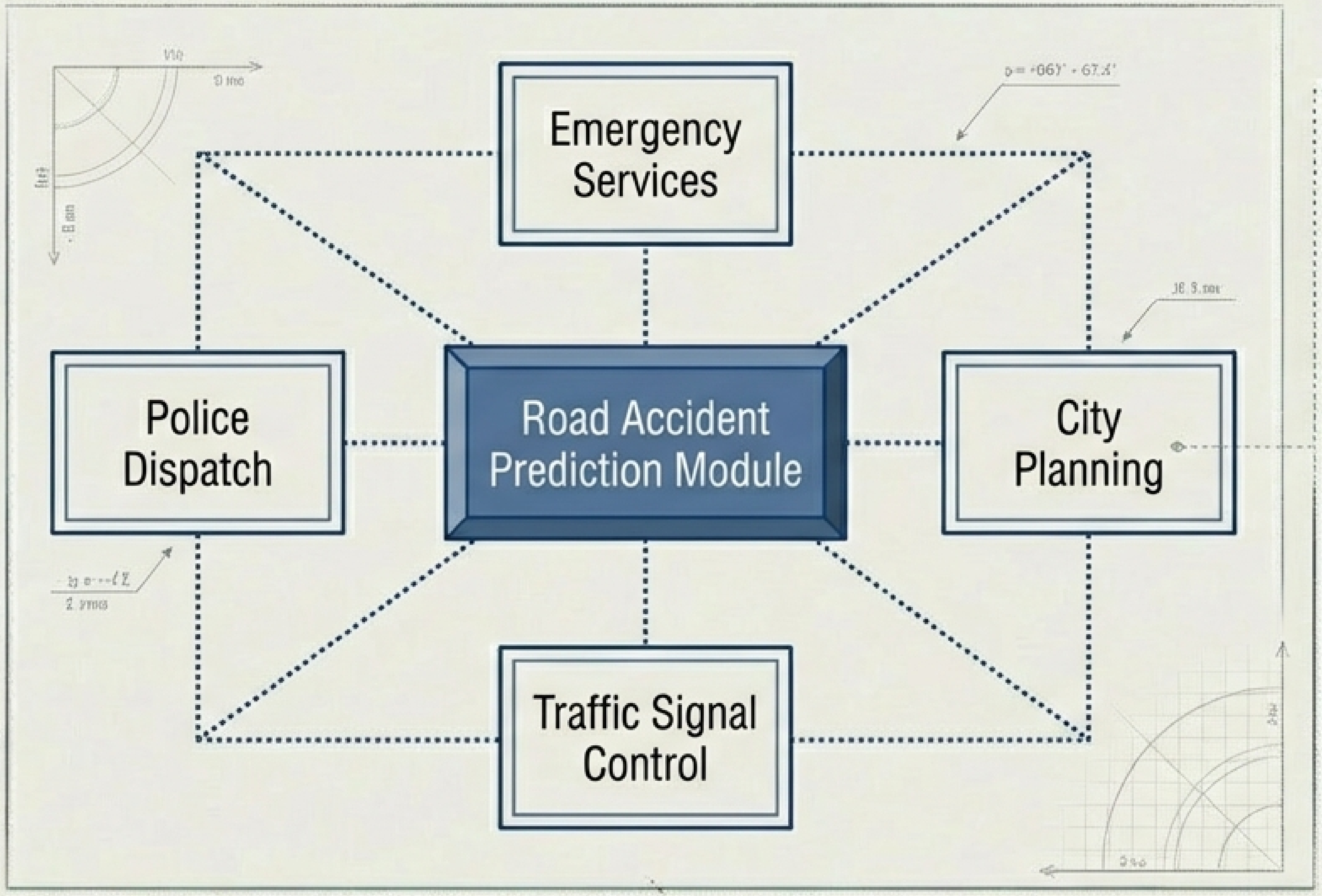


- Supports **data-backed decision-making** rather than than intuition.
- Enables **preventive planning**—allocating resources (police, signage, road repairs) to **high-risk zones** before accidents spike.

Societal Impact

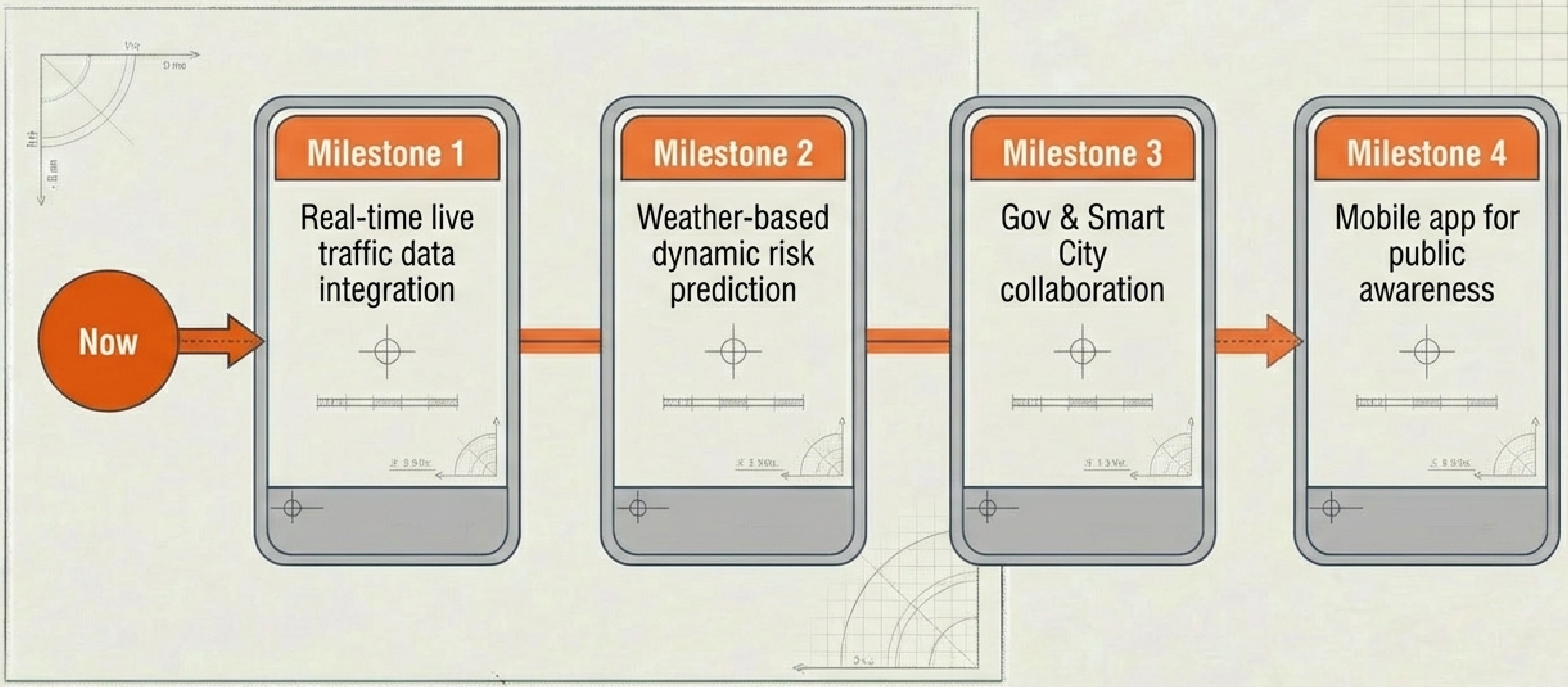
- **Core Goal:** Reduces accident rates through timely intervention.
- **Outcome:** Directly improves public safety in urban environments, creating a safer ecosystem for drivers and pedestrians alike.

Built for Scale



- **Integration**: The architecture is designed to be scalable for smart city integration.
- **Flexibility**: The system is not a standalone silo but a module that can feed into broader municipal management platforms.

Future Scope & Roadmap



FROM HINDSIGHT TO FORESIGHT.

Improving urban safety through intelligent prediction.

Created by Team BACK

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